$\mathscr{F}_t^H$ represents the flow map of Hamiltonian dynamics. Thus $C(0) = 1$. At large τ, if the system is "mixing," and loses memory of its starting point, hence

$$\mathrm{Av}_\beta \left[\boldsymbol{A}(z) \cdot \boldsymbol{B}(\mathscr{F}_\tau^H(z))\right] \approx \mathrm{Av}_\beta \left[\boldsymbol{A}(z)\right] \cdot \mathrm{Av}_\beta \left[\boldsymbol{B}(z)\right]$$

is approximately independent of τ. Thus the correlation should approximately vanish at large time, provided the scalar product of the average of $\boldsymbol{A}$ and the average $\boldsymbol{B}$ vanishes. Computing an autocorrelation function $C_\beta(\tau)$ normally involves performing a stationary thermodynamic calculation in which the observable itself depends on a Hamiltonian dynamics calculation.

There is no reason for taking the averages solely with respect to ρ_β. We could in effect assume any prior distribution for the initial conditions. Our choice of Gibbs statistics in defining autocorrelation is made for purely practical reasons—it simplifies the study of the problem by allowing us to bring to bear the common techniques of thermodynamic sampling. In some cases, the Gibbs weight is ignored and the calculation is performed as an average two point correlation along a single trajectory. In some cases this calculation is performed using a single thermostatted molecular dynamics trajectory. Such a calculation is only justified when the thermostat is "weakly coupled" to the physical system. In this setup, the key to accurately computing averaged dynamics of molecular systems is to use thermostat techniques that minimally perturb the physical dynamics. We will briefly discuss how this may be achieved while maintaining the ergodic property.

8.1 Dynamics-Based Thermostat Techniques

Deterministic trajectories can sometimes be used to sample the canonical measure. A very simple technique of this type is Hamiltonian dynamics itself.

Let a Hamiltonian $H(\boldsymbol{q}, \boldsymbol{p})$ be given and let $\mathcal{L}$ be its Liouvillian

$$\mathcal{L} = \nabla_{\boldsymbol{p}} H \cdot \nabla_{\boldsymbol{q}} - \nabla_{\boldsymbol{q}} H \cdot \nabla_{\boldsymbol{p}}.$$

Recall from Chap. 5 that, as the dynamics automatically preserves functions of H, the Liouvillian satisfies $\mathcal{L}\varphi(H) = 0$. We also know that for Hamiltonian dynamics $\mathcal{L}$ is skew-adjoint, i.e., $\mathcal{L}^\dagger = -\mathcal{L}$ which implies that $\mathcal{L}^\dagger \psi(H) = 0$. Hence any density which is a function of H will be preserved, and this includes the canonical one $\psi(H) = \exp(-\beta H)$.

If we select a set of points at random from the canonical distribution, then initiate trajectories of Hamiltonian dynamics from each of these points, the points will remain Gibbs-distributed over time. If the paths themselves are ergodic on the surfaces of constant energy $H = \mathrm{E}$ then the collection of paths may provide a usable sampling of the canonical distribution. Such a sampling technique relies on having the means of choosing initial points from the canonical distribution; as this is the